Linux SSH Brute Force Detection and Triage Using Splunk

This project demonstrates the creation of a SOC-style monitoring workflow using Splunk Enterprise and Ubuntu Linux. Failed SSH authentication attempts were intentionally generated from a Windows system and forwarded into Splunk for detection, alerting, and triage analysis.

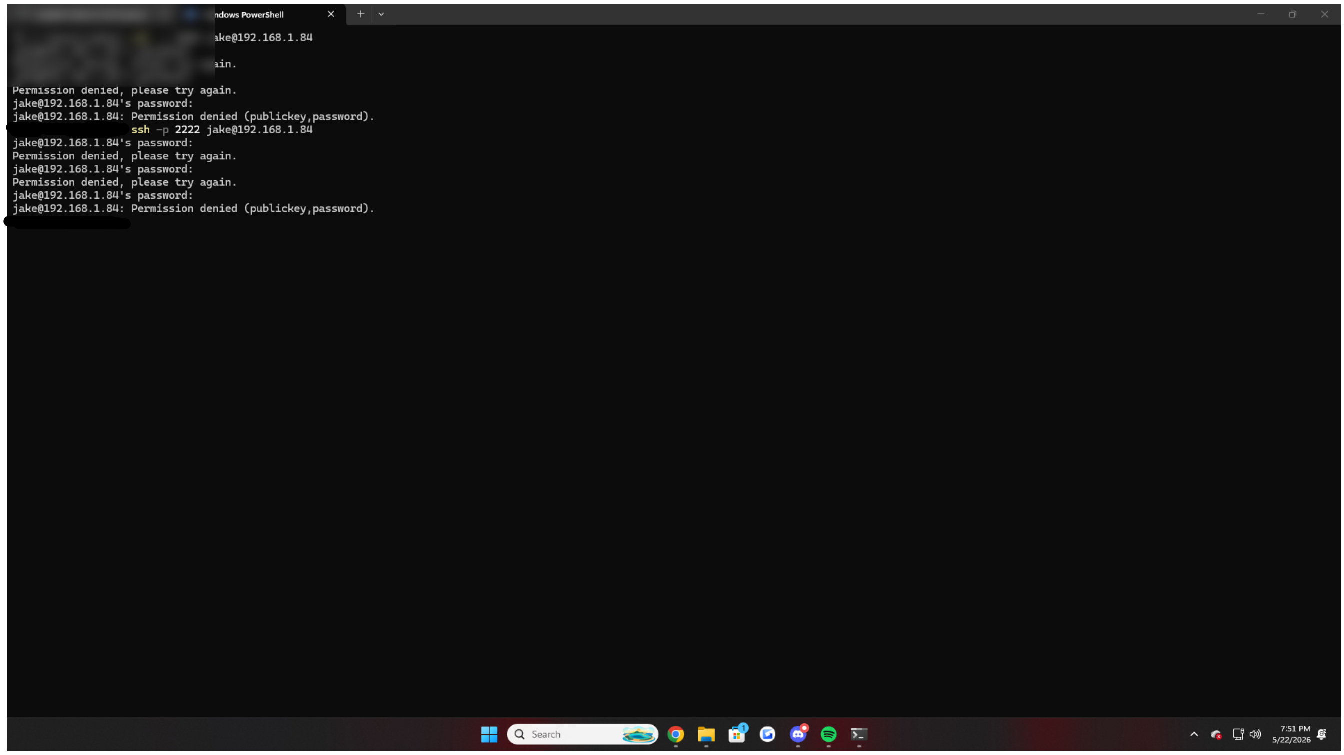
Skills Demonstrated
Splunk Enterprise
Linux Authentication Log Analysis
Security Monitoring and Alerting
SOC Alert Triage
SSH Brute Force Detection
Basic SPL Query Development

Lab Environment

- Ubuntu Linux host configured with SSH access
- Splunk Enterprise running in a Docker container
- Splunk ingesting authentication logs from /var/log/auth.log
- Windows workstation used to simulate failed SSH authentication attempts

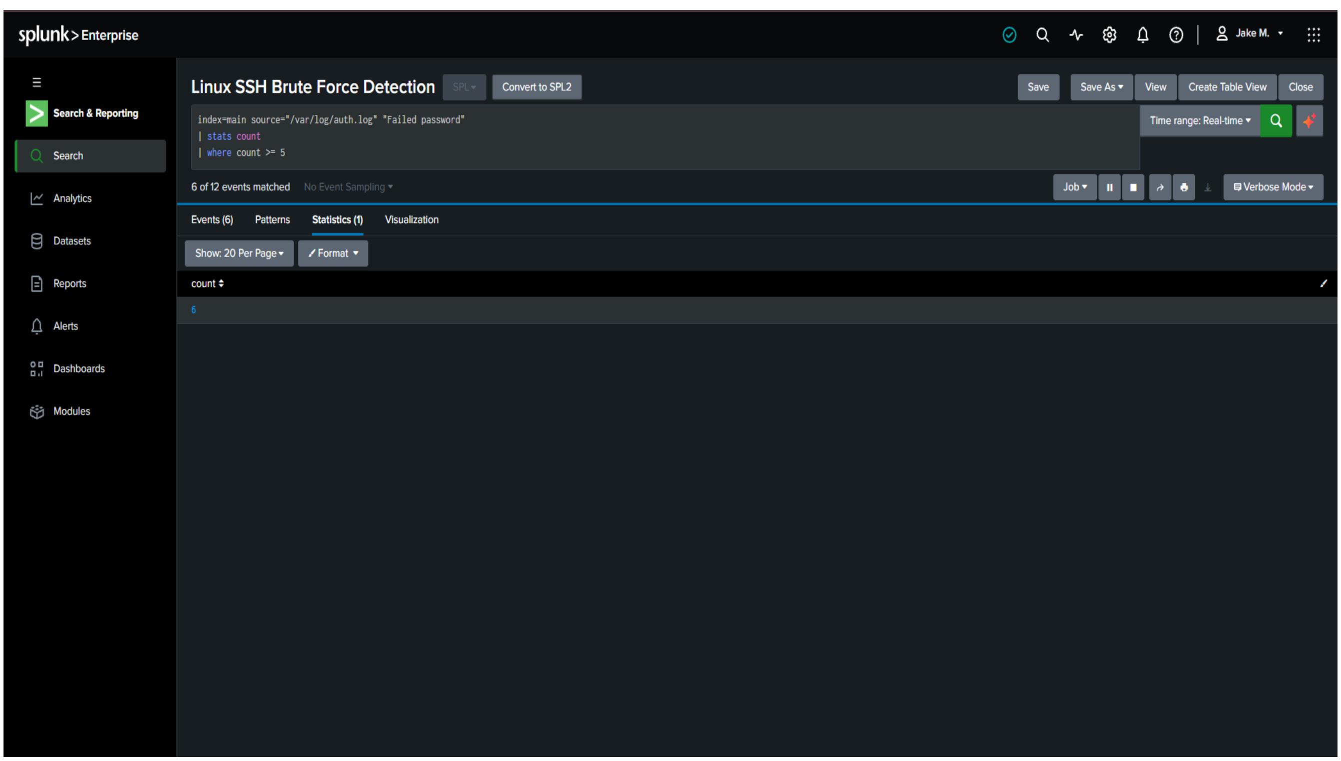
Attack Simulation

Failed SSH login attempts were intentionally generated from a Windows system using PowerShell. Incorrect passwords were entered multiple times to simulate brute-force authentication behavior.



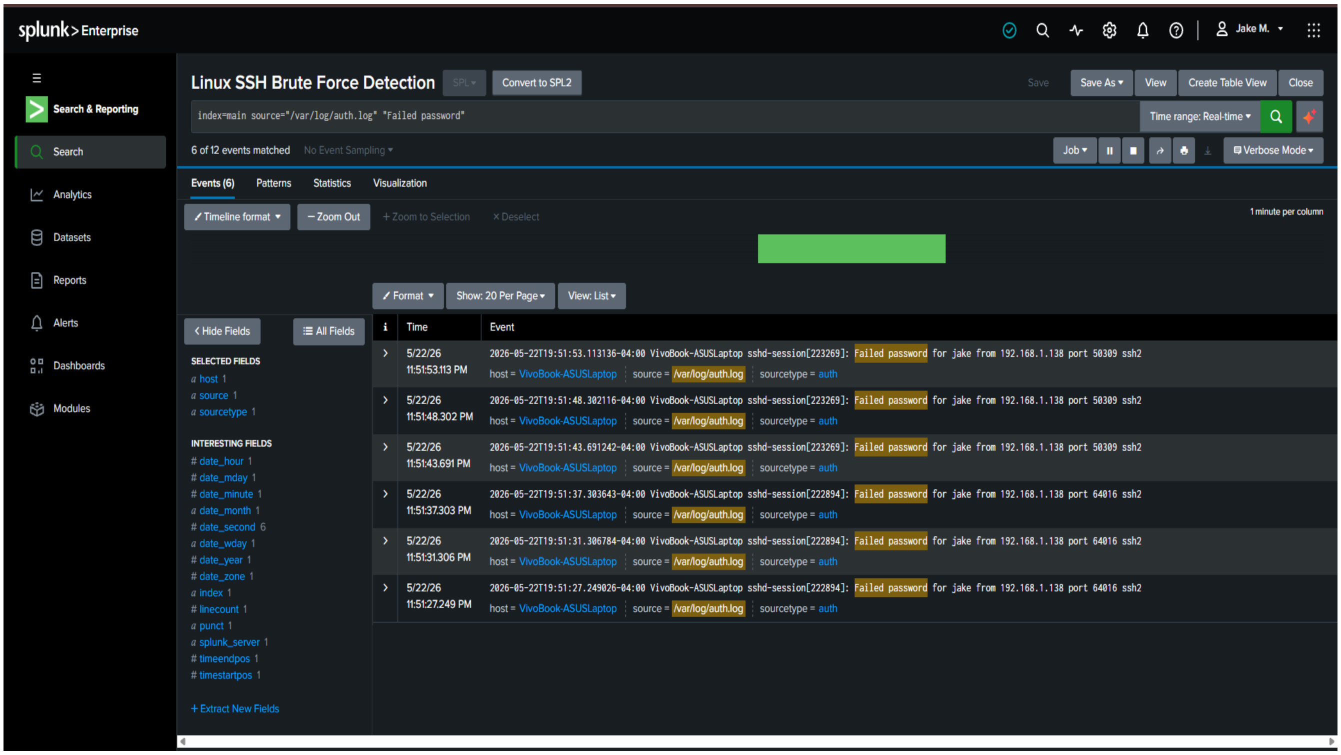
Detection Logic

A Splunk SPL query was developed to identify excessive failed SSH authentication attempts. The detection logic counted matching failed login events and triggered an alert once the threshold exceeded five events.



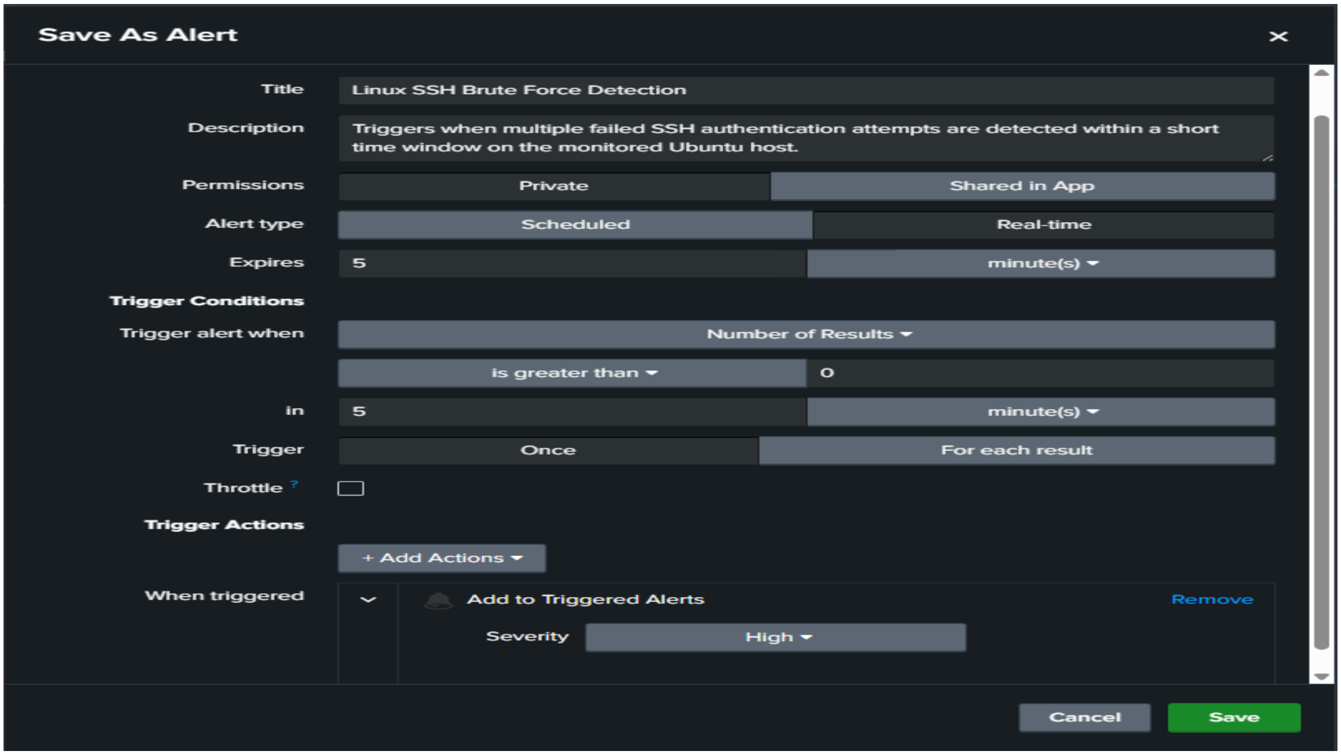
Authentication Log Analysis

Splunk successfully ingested and displayed failed SSH login events from the Ubuntu authentication log. Event analysis included timestamps, usernames, and originating IP addresses.



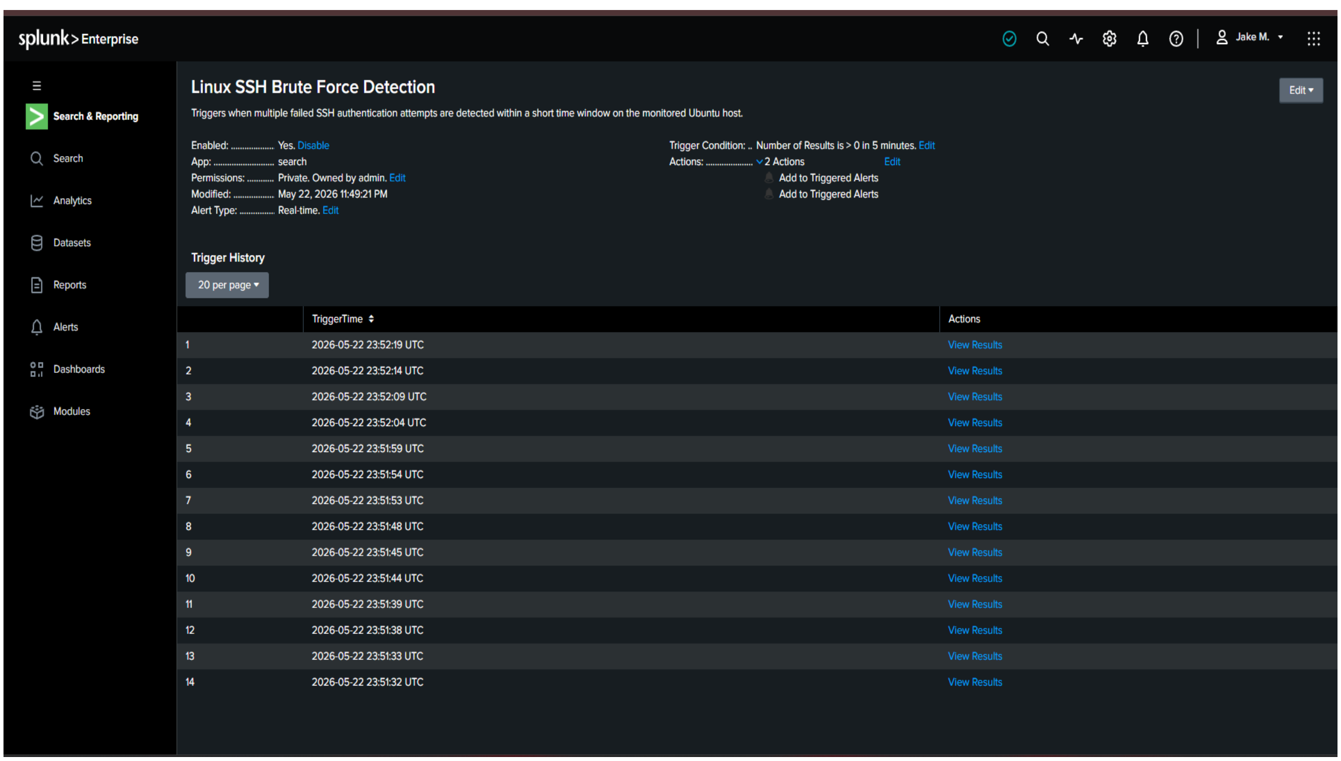
Alert Configuration

A real-time Splunk alert named **Linux SSH Brute Force Detection** was configured to monitor failed SSH authentication attempts occurring within a five-minute window.



Triggered Alert Results

After generating repeated failed login attempts, the Splunk alert triggered successfully. The alert history confirmed multiple detections across the monitored timeframe.



Findings and Conclusion

Splunk successfully detected repeated failed SSH login attempts against the Ubuntu host. The project demonstrated a complete SOC-style workflow including attack simulation, log ingestion, alert creation, detection tuning, and alert triage.

This homelab strengthened practical understanding of:

• Security monitoring workflows
• Linux authentication logging
• Splunk search processing language (SPL)
• Alert configuration and threshold tuning
• SOC investigation and event triage